NOTE The source code for the examples in this chapter is available in the Chapter15/15-begin and Chapter15/15-end folders, containing the initial and final states of the project (https://github.com/ThomasVitale/cloud-native-spring-in-action).

15.1 Deployment pipeline: Acceptance stage

Continuous delivery is one of the fundamental practices we have identified that can support us in our journey to achieve the cloud native goals: speed, resilience, scale, and cost optimization. It's a holistic approach for delivering high-quality software quickly, reliably, and safely. The main idea behind continuous delivery is that an application is always in a releasable state. The primary pattern for adopting continuous delivery is the deployment pipeline, which goes from code commit to releasable software. It should be automated as much as possible and represent the only path to production.

Chapter 3 explained that a deployment pipeline can be composed of three key stages: commit stage, acceptance stage, and production stage. Throughout the book, we have automated the commit stage as a workflow in GitHub Actions. After a developer commits new code to the mainline, this stage goes through build, unit tests, integration tests, static code analysis, and packaging. At the end of this stage, an executable application artifact is published to an artifact repository. That is a *release candidate*.

This section will cover how we can version release candidates for continuous delivery. Then you'll learn more about the acceptance stage, its purpose, and its outcome. Finally, I'll show you how to implement a minimal workflow in GitHub Actions for the acceptance stage. At the end of this stage, the release candidate will be ready to be deployed to production.